AeroSpace Pulsar-1 Personal Orbital Shuttle

Single-Occupant Sub-Orbital & Low-Earth-Orbit Class Vehicle — Model Year 2027

CATALOGUE SKU: **ALZA-SP-PLS1-2027** **1 299 000 000 Kč** vč. DPH

1. Product Overview

The AeroSpace Pulsar-1 is a revolutionary single-person shuttle engineered for high-altitude sub-orbital commutes and low-Earth orbit access. Built around a carbon-titanium composite monocoque chassis, the vessel features an automated AI flight director system, redundant life-support scrubbers, and dual hybrid ion-plasma thrusters. Exclusively distributed via Alza.cz with white-glove orbital delivery options.

2. Technical Specifications

Parameter	Value	Standard / Notes
Crew Capacity	1 Operator (Single cockpit)	Anatomical Zero-G bucket seat
Max Velocity	Mach 24.5 (28,000 km/h)	LEO insertion capability
Dry Mass	1,420 kg	Carbon-mesh composite shell
Propulsion	Hyperion-9 Ion/Plasma Engine	Pulse-detonation RCS thrusters
Life Support (O2)	48 Hours nominal runtime	Dual active lithium scrubbers
Power Source	Solid-State Fusion Cell Mk-IV	750 kWh capacity
Warranty	3 Years Alza Premium Assist	Includes 24/7 Orbital Recovery

> **Safety Notice:** Ensure pilot has completed Alza Flight Level II certification before initiating main thruster priming sequence. Operating without calibrated thermal shield tiles voids warranty.

3. On-Board Diagnostic Matrix & Error Codes

The Pulsar-1 flight controller continuously monitors hull integrity, engine telemetry, and life-support subsystems. Below is the reference table for diagnostic error codes displayed on the main HUD and warning light panel.

Code	System Subroutine	Severity	Description & Recommended Action
ERR-101	O2 Scrubber Flow	CRITICAL	Oxygen recirculation rate dropped below 85%. Switch to auxiliary bottle reserve immediately.
ERR-104	RCS Nozzle Alignment	WARNING	Pitch RCS thruster offset detected (>3.2°). Perform manual recalibration from control panel.
ERR-208	Thermal Shield Grid	CRITICAL	Ablative layer degradation on ventral starboard tiles. Atmospheric re-entry prohibited.
ERR-302	Fusion Core Cell Temp	WARNING	Coolant loop temperature exceeded 420 K. Secondary heatsink fans engaged.
ERR-315	Plasma Injector Pressure	CRITICAL	Fuel manifold pressure drop detected. Engine automatically throttled to idle. Inspect valve V-4.
ERR-401	Navigation / Lidar Sync	INFO	Orbital positioning pulse delayed >120ms. Recalculating orbit trajectory vectors via terrestrial GPS.
ERR-505	Canopy Pressure Seal	CRITICAL	Micro-leakage detected in primary cockpit gasket. Engage pressure suit emergency lock.

4. Service & Maintenance Guidelines

Regular maintenance must be performed every 50 flight hours or 10 orbital re-entries, whichever comes first. Only authorized Alza Aerospace Technicians are certified to replace Fusion Core components and Ablative Heat Shielding. Replacement parts must be selected strictly from the official Pulsar-1 Spare Parts Catalog.

> **Emergency Procedure:** In case of simultaneous ERR-101 and ERR-505 warnings, trigger the emergency escape pod ejection sequence by pulling the dual yellow handles located between your knees.